MILITARY SPECIFICATION

MUZZLE BORESIGHT DEVICE 9364909 (M27 FOR 120MM GUN TUBE)

This Specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements and quality assurance provisions for the cased series of the Muzzle Boresight Device 9364909 (M27 for 120MM Gun Tube), herein referred to as MBD.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense index of Specifications and Standards (DoDISS), and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-F-13926	-	Fire Control Material, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	-	Parts, Equipment, and Tools for Army Material, Packaging and Packing of
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance, and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of solicitation.

PACKAGING DATA SHEET

P9364909 - Packaging of Muzzle Boresight Device (M27 for 120MM Gun Tube)

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

9364909 - Muzzle Boresight Device (M27 for 120MM Gun Tube)
9364930 - Test and Alignment Fixture for M27 Muzzle Boresight Device (120MM)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting acivity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Fabrication. The MBD shall be manufactured in accordance with Drawing 9364909 and drawings pertaining thereto and, when assembled, shall meet the requirements of this specification.

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3.2 General specification. The provisions of MIL-F-13926 apply.

3. 2.1 Ambient conditions. Standard ambient conditions shall be as follows:

a.	Temperature	73° + 18°F
b.	Relative humidity	50 pecent ± 30 percent
c.	Atmosphere pressure	28.5 + 2.0 - 3.0 in Hg.

3.3 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.4 Cleanliness.

a. Defects. There shall be no evidence of moisture, grease, condensates, fractures, and adhesive separations on any glass component when viewing either through the eyepiece or objective end.

b. Dirt. When viewing through the eyepiece, there shall be no more than three (3) particles of dirt or foreign material (classed as dirt) appearing within a 50 Mil diameter central field of view. There shall be no more than three dirt particles appearing outside the specified field of view. The angular separation between dirt particles shall be at least 15 Mils. Dirt particles smaller than 1/4 reticle dot width shall be ignored regardless of distribution. The size of any particle shall not exceed the apparent width of the reticle dot.

3.5 Performance.

3.5.1 Collimation. Collimation error shall not exceed 50 seconds.

3.5.2 Parallax. Maximum allowable parallax shall not exceed 10 seconds.

3.5.3 Collimation repeatability. The collimation repeatability error shall not exceed 30 seconds. All measurements must meet the collimation requirements of 3.5.1.

3.5.4 Transmission measurement. Transmission shall be 60 percent minimum when measured using a photopic filter.

3.5.5 Resolution measurement. The resolution of the system shall not exceed 8 seconds.

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3 .5.6 Eyepiece focus measurement. The eyepiece setting shall be -0.75 ± 0.25 diopters.

3.5.7 Magnification. The requirement for magnification of 7X plus or minus 0.14X.

3.5.8 Field of view (FOV). The FOV shall be 6 degrees minimum.

3.6 Purging. The MBD shall be purged with dry nitrogen for a period of time until it is determined that the dew point is at least -25 degrees F at atmospheric pressure. The unit shall be sealed immediately after purging.

3.7 Sealing. Care will be utilized to keep MBD internal atmosphere purged. The MBD shall show no evidence of leakage in excess of 0.1 PSI when subjected to an internal pressure of 5 PSIG plus or minus 0.1 PSI for a period of one hour. The interior atmosphere of the accepted MBD shall contain nitrogen having a dew point at least as low as minus 25°F at a pressure of between 0.25 and 0.50 PSIG.

3.8 Interchangeability verification. All MBD's shall be interchangeable (see 4.7.6.1) .

3.9 Environmental.

3.9.1 Storage temperature. The MBD shall show no evidence of cement separation, glass breakage, or other physical failure and shall meet the requirements of this specification at standard temperature (plus 15 degrees C to 32 degrees C) after having been exposed and thermally stabilized at temperatures of plus 71 degrees C and minus 62 degrees C.

3.9.2 Operating temperatures. The MBD shall be mechanically operable while thermally stabilized at plus 65 degrees C and minus 40 degrees C. The rate of temperature change shall not exceed 10 degrees C/hour.

3.9.3 Vibration. The cased MBD shall be capable of withstanding vibration when subjected to Test Method 514.2, Procedure X, Curve AV, Equipment Category G, of M1L-STD-810.

3.9.4 Shock. The MBD shall withstand shock in accordance with Test Method 516.2, Procedure I, Figure 516.2-2, Time Duration 11 milliseconds, of MIL-STD-810, except that the peak acceleration shall be 40 g’s.

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3.9.5 Humidity. The MBD shall operate as specified herein at up to 100 percent relative humidity (see 4.7.9.7).

3.9.6 Dust. The MBD shall operate as specified herein after exposure to moving dust for not less than 12 hours at velocities up to 1750 feet per minute (see 4.7.9.8).

3.9.7 Solt fog. The MBD shall operate as specified herein after exposure to a 5 percent sodium chloride atomized spray for not less than 48 hours (see 4.7.9.9).

3.10 Workmanship.

3.19.1 Optical inspection. The MBD shall be viewed with the unaided eye from the objective end against a background of approximately 300 foot candles. There shall be no cracks, chips, or band separation. When a focused target is viewed through the eyepiece end, there shall be no scratches, chips, dirt, or coating deterioration which would be a distraction to the observer.

3.10.2 Mechanical inspection. The mechanical parts of the unit shall be inspected for broken parts, completeness, workmanship and functionability. The locating block shall be checked for freedom and smoothness of operation. Paint chips or nicks which do not affect the function of the instrument shall be touched up. All burrs and excess material shall be-removed. from castings and forgings. All surfaces, including threads, shall be free from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the quality assurance provisions of MIL-F-13926 and the conditions of 3.2.1.

4.4 First article. The requirement for first article approval and the designation of responsibility for first article inspection to either the Government or the contractor shall be specified in the contract.

4.4.1 Sample. The first article sample shall be three assemblies selected at random by the Government representative from the first 15 production assemblies.

4.4.2 Inspection. The sample shall be subjected to all the inspections specified in Table I.

4.4.3 Failure. Failure of any assembly to meet any requirement shall be cause for refusal to grant first article approval. The Government reserves the right to terminate first article inspection upon failure of any assembly to comply with any stated requirement.

4.4.4 Responsibility. The contractor shall inspect the sample for conformance to all contractual requirements and shall submit a record of this inspection with the sample and certificates of conformance for materials. The Government reserves the right to witness inspections performed by the contractor.

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TABLE I. First article tests

Item	Characteristic	Requirement	Test Procedure
1	Fabrication	3.1	Applicable visual drawings
2	General specifications	3.2	MIL-F-13926/Visual
3	Performance	3.5	4.7.4
4	Sealing	3.7	4.7.8
5	Shock	3.9.4	4.7.9.3
6	Vibration	3.9.3	4.7.9.2
7	High temperature	3.9.1 & 3.9.2	4.7.9.4
8	Low temperature	3.9.1 & 3.9.2	4.7.9.5
9	Temperature cycling	3.9.1 & 3.9.2	4.7.9.6
10	Humidity	3.9.5	4.7.9.7
11	Dust	3.9.6	4.7.9.8
12	Salt fog	3.9.7	4.7.9.9

4.5 Quality conformance inspection. Inspection plans A and B of 4.5.1 and 4.5.2 shall apply.

4.5.1 Inspection plan A.

4.5.1.1 General sample and tests. One assembly, as a control sample, shall be selected at random by the Government representative from each 100 assemblies produced and shall be subjected to all the tests in Table III.

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4.5.1.2 Environmental sample and tests. Two assemblies, as a control sample, shall be selected at random by the Government representative from each 100 assemblies produced or from each month's production, whichever occurs first. All the tests in Table IV shall be applied separately to each assembly in the sample.

4.5.1.3 Rejection. If any one assembly of either sample fails to meet any specified requirement, the lot shall be rejected. Rejected lots may be resubmitted in accordance with the provisions of MIL-STD-105.

4.5.2 Inspection plan B.

4.5.2.1 Sample and tests. Each assembly in every lot shall be subjected to all the tests in Table II and shall be examined visually for completeness, improper assembly, and evidence of poor workmanship.

4.5.2.2 Rejection. If any assembly fails to meet any specified requirement, the defective assembly shall be rejected and removed from the lot and may be resubmitted only after all defects have been corrected.

4.5.3 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.4 Packaging inspection. The sampling and inspection of the preservation, packaging, packing, and container marking shall be in accordance with the provisions of Packaging Data Sheet P9364909 and MIL-P-14232.

4.6 Inspection provisions.

4.6.1 Submission of product. Unless otherwise specified by the contracting officer, inspection lot size, lot formation, and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.7 Examination and tests.

4.7.1 Components and subassemblies. All components and subassemblies shall be inspected in accordance with the applicable Quality Assurance Provisions of MIL-F-13926.

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4.7.2 Boresight collimator. Examination and tests herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables II and III herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232, Level B. Tables I, II, and III shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by item or lot.

4.7.3 Inspection characteristics.

4.7.3.1 100 Percent inspection. The following requirements and related tests shall be subjected to 100 percent inspection.

TABLE II. Performance tests

Item	Characteristic	Requirement	Test Procedure
101	Collimation	3.5.1	4.7.4.1
102	Parallax	3.5.2	4.7.4.2
103	Repeatability/collimation	3.5.3	4.7.4.3.2
104	Resolution	3.5.5	4.7.4.5
105	Eyepiece focus	3.5.6	4.7.4.5
106	Optical inspection	3.9.1	4.7.5.2
107	Mechanical inspection	3.9.2	4.7.5.3
108	Interchangeability	3.8	4.7.6
109	Purging	3.6	4.7.7
110	Sealing	3.7	4.7.8

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4.7.3.2 Sampling inspection.

TABLE III. Classification of defects

Class
Critical: None
Major: AQL 0.65% Defective

Item	Characteristic	Requirement	Test Procedure
201	Magnification	3.5.7	4.7.4.6
202	Field of View (FOV)	3.5.8	4.7.4.7
203	Transmission	3.5.4	4.7.4.4

Minor: None

NOTE: Tests in Table II and III shall be conducted at an ambient temperature within the bounds of 3.2.1.

TABLE IV. Environmental tests

Item	Characteristic	Requirement	Test Procedure
301	Storage temperature	3.9.1	4.7.9.1
302	Operating temperature	3.9.2	4.7.9.1
303	Vibration	3.9.3	4.7.9.2
304	Shock	3.9.4	4.7.9.3
305	High temperture	3.9.1 & 3.9.2	4.7.9.4
306	Low temperature	3.9.1 & 3.9.2	4.7.9.5
307	Temperature cycling	3.9.1 & 3.9.2	4.7.9.6

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4.7.4 Methods of inspection.

4.7.4.1 Collimation test method. Collimation error shall be measured by viewing through the eyepiece of the MBD and comparing the coincidence of the reticle boresight referenced to the projected reticle image of the Collimation/Parallax test fixture. The accuracy is determined by comparing the MBD reticle reference point to the projected reticle pattern. Collimation is met when collimation error is within 50 seconds.

4.7.4.2 Parallax test method. Parallax shall be measured by viewing through the eyepiece of the MBD and observing the apparent displacement between the MBD reticle and the projected reticle image of the Collimation/Parallax test fixture as the observer moves his eye from side to side or up and down. The accuracy is determined by comparing the displacement to the reticle pattern marking. Maximum allowable parallax error is 10 seconds.

4.7.4.3 Repeatability/collimation.

4.7.4.3.1 Repeatability/collimation test fixture. The collimation/parallax test fixture or gun tube is to be used for the image repeatability test.

4.7.4.3.2 Repeatability/collimation measrement. The collimation error is noted with the MBD inserted in the Collimation/ Parallax Test Fixture. Unit will be withdrawn then reinserted into gun tube or test fixture. The collimation error is noted and comapred to the initial reading. Repeatability within 30 seconds is acceptable. Both measurements must meet the collimation requirement of 3.5.1.

4.7.4.4 Transmission.

4.7.4.4.1 Transmission test fixture. The test fixture consists of a light source and integrating sphere, mount and calibrated telephotometer.

4.7.4.4.2 Transmission measurement. The telephotometer is set to view the light emitted from the light source and normalized. Light from the light source is back lighted through the eyepiece of the MBD. When the MBD is placed in the light path it shall be positioned so that light exit of the light source is at the focal point of the MBD's eyepiece. The telephotometer is moved to view the objective of the MBD and the transmission is measured. Transmission shall be 60 percent minimum when utilizing a photopic filter.

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4.7.4.5 Resolution/eyepiece focus measurements.

4.7.4.5.1 Resolution/eyepiece focus test equipment. A four (4) quadrant eight (8) second pattern on the collimator reticle must be discernible when viewed through the eyepiece of the MBD with the aid of a dioptometer.

4.7.4.5.2 Resolution/eyepiece focus measurement. A dioptometer setting of -0.75 plus or minus 0.25 diopters must register when best focus of the reticle is obtained when viewed through the eyepiece with the aid of a dioptometer.

4.7.4.6 Magnification. The magnification is measured by dividing the objective clear aperture, or entrance pupil, by the diameter of the exit pupil. The exit pupil diameter is measured with a calibrated dynameter. The requirement for magnification is 7X plus or minus 0.14X.

4.7.4.7 Field of view (FOV). The FOV is measured by observing the minimum separation, S, on a target at a distance, $_D$, from the MBD. FOV = 2 arc tan (S/2D).

4.7.5 Mechanical.

4.7.5.1 Optical/mechanical workmanship.

4.7.5.2 Optical/inspection. The MBD shall be viewed with the naked eye from the objective end against a background of approximately 300 foot candles. There shall be no cracks, chips, or bond separation. When a focused target is viewed through the eyepiece end, there shall be no scratches, chips, dirt, or coating deterioration which would be a distraction to the observer. The presence of an anti-reflective coating, where required, shall be verified during in-process inspection.

4.7.5.3 Mechanical inspection. The mechaniscal parts of the MBD shall be inspected for broken parts completeness workmanship and functionability. The locating block shall be checked for freedom and smoothness of operation. Paint chips or nicks which do not effect the function of the instrument shall be touched up. All burrs and excess material shall be removed from castings and forgings. All surfaces, including threads, shall be free from burrs and sharp edges.

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4.7.6 Interchangeability.

4.7.6.1 Interchangeability fixture. The ability to insert the MBD into the test fixture P/N 9364930 barrel or gun tube confirm's the interchangeability of the device.

4.7.7 Purging. This portion of the test shall be performed utilizing dew point tester, Type 7000 or equivalent. The MBD shall be purged with dry nitrogen for a period of time until it is determined that the dew point is at least -25 degrees at atmospheric pressure. Suitable valve adapters shall be used to facilitate this test. The unit shall be sealed immediately after purging.

4.7.8 Sealing. This test shall be performed utilizing a suitably calibrated pressure gage. Suitable valve adapters shall be used to facilitate the test. Care will be utilized to keep the MBD internal atmosphere purged. The MBD shall be pressurized with dry nitrogen to 5 plus or minus .1 PSI for a minimum of one hour. There shall be no evidence of leakage in excess of 0.1 PSI. Prior to shipment, the unit will be sealed at a pressure of between 0.25 and 0.50 PSIG.

4.7.9 Environmental sampling. Three MBD's shall be selected at random from each 50 produced, or from each month’s production whichever occurs first. Each sample shall meet the requirements and tests in Table IV.

NOTES: Prior to environmental, the MBD shall have met the examination and tests in Table II.

4.7.9.1 Storage and operating temperature. The test equipment utilized in this test shall be in accordance with “Test Facilities” requirements of MIL-F-13926. The uncased MBD shall be placed in the test chamber and the temperature of the test chamber raised gradually (rate of temperature change in the test chamber shall not exceed 10 degrees C per hour throughout the temperature cycling test) to 65 degrees C. The MBD shall remain at this temperature for a minimum of three hours. At the completion of this period and while at 65 degrees C, the MBD locating block shall be depressed three times and observed to return to its original position each time. Upon completion of this test, the temperature of the test chamber shall be gradually raised (see above rate of temperature change) to 7.1 degrees C.

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The MBD shall remain at this temperature for a minimum period of three hours. The temperature of the chamber shall be gradually reduced (see above rate of temperature change) to minus 40 degrees C. The MBD shall remain at this temperature for a minimum of three hours.

At the completion of this period and while at minus 40 degrees C, the MBD locating block shall be depressed three times and observed to return to its original position each time. Failure of the locating block to return to its original position at the operating temperature, minus 40 degrees C and plus 65 degrees C, constitutes failure of the test. Upon completion of this test, the temperature of the test Chamber shall be gradually reduced (see above rate of temperature change) to minus 62 degrees C. The MBD shall remain at this temperature for a minimum period of three hours.

The temperature chamber shall be gradually raised (see above rate of temperature change) to ambient temperature plus 15 degrees to plus 32 degrees C. NoteL Tolerance on constant temperature is plus or minus 2.8 degrees C. The MBD shall be removed and subjected to the examination and tests in Table II.

4.7.9.2 Vibration (secured cargo). The cased MBD shall be vibrated in accordance with Test Method 514.2, Procedure X, Curve AV, Equipment Category G, of MIL-STD-81O. Vibration shall be applied along each of three mutually perpendicular axes of the MBD. Vibrating equipment capable of providing the test level and frequency ranges specified in the above test method shall be utilized for this test. Unless otherwise specified, the test equipment shall be in accordance with the “Test Facilities" requirements of MIL-F-13926. The MBD shall be removed and subjected to the examination and tests in Table II.

4.7.9.3 Shock. The MBD shall be shocked in accordance with Test Method 516.2, Procedure I, Figure 516.2-2, Time Duration 11 milliseconds, of MIL-STD-81O, except that the peak acceleration shall be 40 g's. The MBD shall be removed from the shock fixture and subjected to the examination and tests in Table II.

4.7.9.4 High temperature. Subject the MBD to the high temperature test specified in MIL-STD-81O, Method 501.1, Procedure I. The highest operating temperature shall be plus 71 degrees C. The performance tests of Table II shall be applied.

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4.7.9.5 Low temperature. Subject the MBD to the Low Temperature test specified in MIL-STD-810, Method 502.1, Procedure I. The storage temperature shall be minus 62 degrees C. The lowest operating temperature shall be minus 40 degrees C. The performance tests of Table II shall be applied.

4.7.9.6 Temperature cycling. The rate of temperature change in the test chamber shall not exceed 10 degrees C per hour throughout the temperature cycling test of 4.7.10.1. The performance tests of Table II shall be applied.

4.7.9.7 Humidity. Place the MBD in a humidity chamber and subject it to the humidity test specified in MIL-STD-81O, Method 507.1, Procedure IV. During Steps 4, 6, and 8 of Procedure IV, subject the MBD to the performance tests of 4.7.4. After exposure, subject the assembly to the performance tests of Table 11.

4.7.9.8 Dust. Subject the MBD to the dust test specified in MIL-STD-81O, Method 509.1, Procedure I. At the completion of this test, subject the MBD to the performance tests of Table II.

4,7,10.9 Salt fog. Subject the MBD to the corrosion test specified in MIL-STD-81O, Method 509.1, Procedure I. At the conclusion of this test, subject the MBD to the performance tests of Table II.

5. PACKAGING

5.1 Packaging, packing, and marking. Packaging, packing, and marking shall be in accordance with Packaging Data Sheet P9364909. The level of protection shall be as specified in the procurement document (see 4.1.3).

6. NOTES

6.1 Intended use. The MBD 9364909 is specifically used for precise sight alignment facilities required by modern high performance gun barrels.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable levels of preservation, packaging, and packing.
c. Applicable packaging data (see 5.1).
d. Applicable stock number.
e. Provisions for first article testing.

6.3 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by, and identified as, ARRADCOM, Edgewood Arsenal drawing, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data cognizance of ARK.

Custodian:
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(Project 1240-A839)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-48640 (AR)	MUZZLE BORESIGHT DEVICE 9364909 (M27 FOR 120MM GUN TUBE)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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INCH-POUND

MIL-M-48640 (AR)
AMENDMENT 1
29 April 1991

MILITARY SPECIFICATION

MUZZLE BORESIGHT DEVICE: 9364909
(M27 FOR 120MM GUN TUBE)

This amendment forms a part of MIL-M-48640(AR), dated 31 January 1986, and is approved for use by the U.S. Army Armament, Munitions, and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 3

3.5.1, delete in its entirety and substitute:

"3.5.1 Collimation. Collimation error, defined as the angular difference in the alignment of the optical axis with the mechanical axis, shall not exceed 20 seconds when the Muzzle Boresight Device is mounted in the test fixture."

3.5.3, delete in its entirety and substitute:

"3.5.3 Collimation repeatability. The collimation repeatability error shall not exceed 15 seconds. All measurements must meet the collimation requirements of 3.5.1."

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4.7.4.1, Delete last sentence in its entirety and substitute:

"Collimation is met when collimation error is within 20 seconds as required by paragraph 3.5.1, when rotated through 360", minimum of four readings 90° apart".

4.7.4.3.2. delete fourth sentence in its entirety and substitute:

"Repeatability within 15 seconds is acceptable."

Custodian
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Preparing Activity:
Army-AR
(Project 1240-AB15)

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